

Philip B. Gottschalk

Software Engineer | Graphics, Rendering & Apple Platforms

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Professional Profile

I'm a seasoned software engineer with 25+ years' development experience specialising in graphics, rendering, custom UI systems and Apple-platform software. I'm a veteran in Objective-C, Metal and native macOS/iOS development, with current hands-on experience in Swift, SwiftUI, and SwiftData.

Technical Expertise

Languages

- Objective-C, Swift, C++, Metal Shading Language

Apple

- iOS, macOS, SwiftUI, UIKit, AppKit, Metal, Core Graphics, Core Image, Core Text, Core Animation, CloudKit

Graphics

- GPU rendering, Metal render/compute pipelines, shaders, custom rendering engines, texture management, real-time rendering

Engineering

- Architecture, custom UI systems, multithreading/concurrency, memory optimization, performance profiling, debugging

Tools

- Xcode, Instruments, XCTest, Git, App Store Connect

Recent Software Development

HowLS — Swift / SwiftUI | 2026

Productivity application designed to help users manage recurring low-priority tasks without traditional deadline pressure.

- Designed and developed independently from concept through App Store release.
- Completed design, implementation, testing, visual assets and App Store materials in under two weeks.

Block Ed — Objective-C / Metal | 2025

Universal macOS and iOS application developed over approximately two years.

- Designed and implemented the application architecture, UI and rendering systems.
- Developed custom Metal shader effects and GPU render/compute pipelines.
- Built PDF export, undo/redo, CloudKit sync, autosave and backup systems.
- Adapted the interface across macOS, iPhone and iPad, localised into four languages.
- Managed the complete development and App Store release process.

Other 2026 releases

- *Q-Convert 2* (Swift/SwiftUI), *Colour Well* (Swift/Quartz)

Professional Experience

Independent Software Engineer / Developer | 2002–Present

- Long-term independent software development encompassing applications, graphics technologies, developer tools and reusable libraries.
- Designed and implemented software from concept through release, covering architecture, UX, implementation, testing, optimization and maintenance.
- Developed software across macOS, iOS and Windows, including custom graphics, typography, UI, document-generation and file-management technologies.

thePit.com — Chief Web Architect | c. 2000–2001

- Designed the HTML and underlying ASP architecture for the original website and developed a Java stock ticker.

Sanwa International Ltd — Risk Management Analyst / Software Developer | 1998–2000

- Developed *Genesis*, praised as ‘a revelation’ by *Futures & OTC World* (April 1999), within the London branch of the bank’s risk-management department.
- Initially developed as sole programmer to replace static Excel graphs with an interactive graphical risk-management interface.
- Evolved into a database-driven system using SQL Server, with graphs generated dynamically from daily risk data.
- Rolled out to Sanwa branches in Hong Kong, New York and Tokyo.

Selected Technical Projects

Advanced Typography Library

- Custom GPU-based text-rendering system supporting gradients, shadows, animated texture fills and 3D extrusion, with reusable rendering infrastructure for integration into other applications.

Video Generation Engine

- Automated multilingual promotional and tutorial video generation from plain-text scripts, including screenshot capture, asset integration and video assembly.

Logo Design & Export Tool

- Metal compute-based real-time logo design and asset-generation tool, automatically exporting multiple variants and application configuration files.

Other Software

- Earlier work includes C++ desktop applications, games, backup software, and reusable graphics, typography, cryptography and document-generation libraries.

Education

University of York — BA Mathematics, First Class Honours | 1997